ASEN 6062

Hwk #1

Nicole James

1) Pollard Ch. 1

3.3) given: $\vec{a} = \vec{r}, \vec{b} = \vec{v}$

$(\vec{a}\cdot\vec{b})^2 + (\vec{a}\times\vec{b})^2 = a^2b^2$

show: $v^2 = \dot{r}^2 + \frac{c^2}{r^2}$

$$(\vec{r}\cdot\vec{v})^2 + (\underbrace{\vec{r}\times\vec{v}}_{\vec{c}})^2 = r^2v^2$$

$$(\vec{r}\cdot\vec{v})^2 + \vec{c}\cdot\vec{c} = r^2v^2$$

$$(\vec{r}\cdot\dot{\vec{r}})^2 + c^2 = r^2v^2$$

$$r^2\dot{r}^2 + c^2 = r^2v^2$$

$$\boxed{v^2 = \dot{r}^2 + \frac{c^2}{r^2}}$$

In physical terms, $\dot{r}$ = radial velocity

$\frac{c^2}{r^2}$ = tangential velocity

$T = U + h$ $\qquad$ $\frac{1}{2}v^2 = f_1(r) + h$

$$\frac{1}{2}(\dot{r}^2 + \frac{c^2}{r^2}) = f_1(r) + h$$

$$\frac{1}{2}(r\dot{r}^2 + c^2) = r^2[f_1(r) + h]$$

$$\boxed{r\dot{r}^2 + c^2 = 2r^2[f_1(r) + h]}$$

4.1) show: if $0 < e < 1$ or $e > 1$

$$\mu a|e^2 - 1| = c^2$$

$$a = \frac{r_p + r_a}{2}$$

$a = \frac{r_p + r_a}{2}$

$r = \frac{c^2/\mu}{1+e\cos f} = e\left(\frac{c^2}{\mu e} - r\cos f\right)$

$r_p = \frac{c^2/\mu}{1+e} \qquad r_a = \frac{c^2/\mu}{1-e}$

$a = \frac{1}{2}\left[\frac{c^2/\mu}{1-e} + \frac{c^2/\mu}{-(1+e)}\right]$

$2a\mu = \frac{-c^2(1+e+1-e)}{(1-e)(1+e)}$

$\cancel{2}a\mu = -\frac{\cancel{2}c^2}{1-e^2}$

$\boxed{a\mu(e^2-1) = c^2}$

5.3) given: $A_{ell} = \pi a^2(1-e^2)^{1/2}$ $\quad \dot{A} = c/2 \quad$ if $c \neq 0$

show: if $\underbrace{0<e<1}_{\text{ellipse.}}$ $\quad P = \left(\frac{2\pi}{\sqrt{\mu}}\right)a^{3/2}$

$A = c/2\, P$

$P = \frac{2A}{c} = \frac{2\pi a^2(1-e^2)^{1/2}}{c}$ $\qquad c^2 = \mu a|e^2-1|$

$\qquad\qquad\qquad\qquad\qquad\qquad c^2 = \mu a|-(1-e^2)|$

$= \frac{2\pi a^2 \cancel{(1-e^2)^{1/2}}}{\sqrt{\mu a\cancel{|1-e^2|}}}$

$= \frac{2\pi a^{3/2}}{\sqrt{\mu}}$

$\boxed{P = \frac{2\pi a^{3/2}}{\sqrt{\mu}}}$

5.4) define moment of Inertia

$2I = mr^2$

prove $\ddot{I} = 2T - U = T + h = U + 2h,$

$I = \frac{1}{2}mr^2 = \frac{1}{2}m\vec{r}\cdot\vec{r}$

$$I = \frac{1}{2}mr^2 = \frac{1}{2}m\,\vec{r}\cdot\vec{r}$$

$$\dot{I} = \frac{1}{2}m\left(\vec{r}\cdot\dot{\vec{r}} + \dot{\vec{r}}\cdot\vec{r}\right)$$

$$= m\,\vec{r}\cdot\dot{\vec{r}}$$

$$\ddot{I} = m\left(\dot{\vec{r}}\cdot\dot{\vec{r}} + \vec{r}\cdot\ddot{\vec{r}}\right)$$

$$= m\left(\dot{r}^2 + \vec{r}\cdot\ddot{\vec{r}}\right) \qquad \ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r}$$

$$= \underbrace{mv^2}_{} + \vec{r}\cdot m\left(-\frac{\mu}{r^3}\vec{r}\right)$$

$$= 2T + \underbrace{m\left(-\frac{\mu}{r^3}r^2\right)}_{U}$$

$$\ddot{I} = 2T - U = 2T - (T - h_1)$$

$$= 2U + 2h_1 - U =$$

$$\boxed{\ddot{I} = U + 2h_1 = T + h_1}$$

8.3) given: u = eccentric anomaly
f = true anomaly

$r = a(e\cosh u - 1)$
$n(t-T) = e\sinh u - u$ (8.2) $h>0$

$r = a(1-e\cos u)$
(mean anomaly) $n(t-T) = u - e\sin u$ (8.3) $h<0$

$r = \dfrac{a|e^2-1|}{1+e\cos f}$ (8.5)

$h<0$ (elliptical)

trig identity: $\tan^2\frac{u}{2} = \dfrac{1-\cos u}{1+\cos u}$

$$\frac{\cancel{a}|e^2-1|}{1+e\cos f} = \cancel{a}(1-e\cos u)$$

$$|e^2-1| = (1-e\cos u)(1+e\cos f)$$

$$|e^2-1| = 1 + e\cos f - e\cos u - e^2\cos u\cos f$$

$$r = \left.\frac{a(1-e^2)}{1+e\cos f}\right\} \text{conic eqn}$$

$$a\cos u = ae + r\cos f$$

$$\cancel{a}\cos u = \cancel{a}e + \frac{\cancel{a}(1-e^2)\cos f}{1+e\cos f}$$

$$\cos u = e(1+e\cos f) + (1-e^2)\cos f$$
$$= \frac{e+\cos f}{1+e\cos f}$$

$$\tan^2\frac{u}{2} = \frac{1-\cos u}{1+\cos u} = \frac{1-\dfrac{e+\cos f}{1+e\cos f}}{1+\dfrac{e+\cos f}{1+e\cos f}}$$

$$= \frac{\dfrac{1+e\cos f - e - \cos f}{\cancel{1+e\cos f}}}{\dfrac{1+e\cos f + e + \cos f}{\cancel{1+e\cos f}}}$$

$$= \frac{1+e\cos f - e - \cos f}{1+e\cos f + e + \cos f}$$

$$\tan^2 f = \frac{1-\cos f}{1+\cos f}$$

$$\tan^2 u = \frac{(1-\cos f) + e(\cos f - 1)}{(1+\cos f) + e(\cos f + 1)}$$

$$= \frac{(1-\cos f)(1-e)}{\underbrace{(1+\cos f)}_{\tan^2 \frac{f}{2}}(1+e)}$$

$$\tan^2 \frac{u}{2} = \frac{1-e}{1+e} \tan^2 \frac{f}{2}$$

$$\boxed{\tan \frac{f}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{u}{2} ; h<0}$$

for $h>0$ (hyperbolic)
use same process but for cosh/tanh

$$\boxed{\Rightarrow \tan \frac{f}{2} = \sqrt{\frac{1+e}{1-e}} \tanh \frac{u}{2}, h>0}$$

8.4 show solutions are unique

$0 < e \leq 1$ elliptical

$\underbrace{n(t-T)}_{\text{Mean anomaly}} = u - e\sin u$

$f(u) = u - e\sin u$

$f'(u) = 1 - e\cos u$ $\quad \cos u \in [-1,1]$

$e\cos u|_{max} = e(1) \leq 1$

$f'(u) = 1 - e \geq 0 \; \forall u \Rightarrow$ function is monotonically increasing so only 1 solution for given t.

$e \geq 1$

$f(u) = e\sinh u - u$

$f'(u) = e\cosh u - 1$ $\quad \cosh(0) = 1$ $\quad \cosh u > 1 \; \forall u \neq 0$

$e\cosh u \geq \cosh u \geq 1$

$e\cosh u - 1 \geq 0 \quad \forall u \Rightarrow$ function is monotonically increasing

10.1 (10.2) $l = u - e\sin u$ $\quad 0 < e < 1$

(10.1) $r = a(1 - e\cos u)$

$u - l = -e\sin u$

periodic: $u(l) - l$

(eccentric anomaly - mean anomaly)

$u - l = e\sin u$

$u - (l + 2\pi) = e\sin u$

$\boxed{u - l = e\sin u + 2\pi = e\sin u}$

odd: $u(-l) = -u(l)$ $\qquad -l = -(u - e\sin u)$

$-l = -(u - e\sin u)$

$\boxed{u - l = -e\sin u}$

vanishes: $l = 0$, $l = \pi$

$u - 0 = e\sin u$

$\boxed{u = e\sin u \Rightarrow u = 0}$

$u - \pi = e\sin u$

$\boxed{u = e\sin u + \pi \Rightarrow u = \pi}$

cont. derivative

$\frac{dl}{dl} = \frac{d}{dl}(u - e\sin u)$

$1 = \frac{du}{dl} - e\cos u\frac{du}{dl}$

$dl = (1 - e\cos u)du$

$\frac{du}{dl} = \frac{1}{1 - e\cos u}$

$\frac{d(u-l)}{dl} = \frac{du}{dl} - 1 = \frac{1}{1 - e\cos u} - 1$

$= \frac{1 - (1 - e\cos u)}{1 - e\cos u}$

$\frac{d(u-l)}{dl} = \frac{e\cos u}{1 - e\cos u} \to \frac{\text{continuous}}{\text{continuous} \,\&\, \neq 0}$

$u(l) - l = \sum_{n=1}^{\infty} u_n \sin nl$

prove: $u_n = \frac{2}{\pi n}\int_0^{\pi} \cos n(u - e\sin u)du$

fourier coefficients:

$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)dx$

$u_n = \frac{2}{\pi}\int_0^{\pi} (u(l) - l)(\sin nl)dl$

(integration by parts) $f = u(l) - l$ $\qquad g = -\frac{1}{n}\cos nl$

$f' = \frac{du}{dl} - 1$ $\qquad g' = \sin nl\, dl$

$\frac{dl}{du} = 1 - e\cos u$

$\frac{du}{dl} \quad \frac{1}{1 - e\cos u}$

$$\frac{dl}{du} = 1 - e\cos u$$

$$\frac{du}{dl} \quad \frac{1}{1-e\cos u}$$

$$f' = \frac{1}{1-e\cos u} - 1$$

$$f' = \frac{e\cos u}{1-e\cos u}$$

$$f'dl = \frac{e\cos u}{\cancel{1-e\cos u}}\cancel{(1-e\cos u)}du$$

$$f'dl = e\cos u\,du$$

$$u_n = \frac{2}{\pi n}\int_0^\pi \cos nl\, f'$$

$$= \frac{2}{\pi n}\int_0^\pi \cos nl\; e\cos u\,du$$

$$= \frac{2}{\pi n}\int_0^\pi \frac{\cos(u - e\sin u)}{\cancel{e\cos u}}\,\cancel{e\cos u}\,du$$

$$\boxed{u_n = \frac{2}{\pi n}\int_0^\pi \cos(u - e\sin u)\,du}$$

10.2 given: Q_0, Q_1 pts on elliptical orbit
u_0, u_1 eccentric anomalies
$u_1 > u_0$

prove: distance Q_0Q_1 is $2a\sin\alpha\sin\beta$

$\alpha = \frac{1}{2}(u_1 - u_0) \qquad \cos\beta = e\cos\frac{1}{2}(u_1+u_0)$

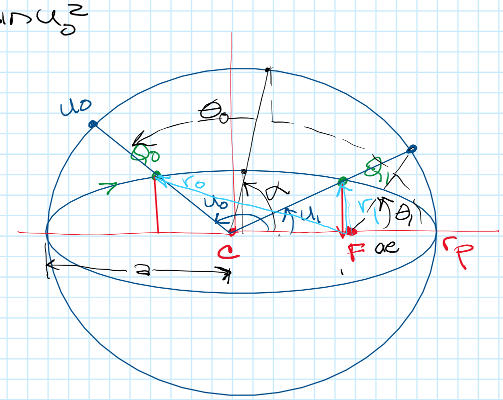
$0 < \beta < \pi \qquad \gamma = \frac{1}{2}(u_1+u_0)$

$$\vec{r} = (a\cos u - ae,\ a\sqrt{1-e^2}\sin u)$$

$$|\vec{r}_1 - \vec{r}_0|^2 = a^2(\cos u_1 - \cos u_0)^2 - a^2(1-e^2)(\sin u_1 - \sin u_0)^2$$

$$(\cos u_1 - \cos u_0)^2 = \cos u_1^2 - 2\cos u_0\cos u_1 + \cos u_0^2$$

$$(\sin u_1 - \cos u_0)^2 = \sin u_1^2 - 2\sin u_0\sin u_1 + \sin u_0^2$$



$$\frac{|r_1 - r_0|^2}{a^2} = \cos^2 u_1 + \cos^2 u_0 - 2\cos u_0\cos u_1$$
$$- (1-e^2)(\sin^2 u_1 + \sin^2 u_2 - 2\sin u_0\sin u_1)$$

$$\cos u_1 - \cos u_0 = -2\sin\frac{u_1+u_0}{2}\sin\frac{u_1-u_0}{2}$$

$$\sin u_1 - \sin u_0 = 2\cos\frac{u_1+u_0}{2}\sin\frac{u_1-u_0}{2}$$

$$\frac{|\vec{r}_1 - \vec{r}_0|^2}{a} = 4\sin^2\alpha\sin^2\gamma + 4(1-e^2)\cos^2\gamma\sin^2\alpha$$

$$= 4\sin^2\alpha(\sin^2\gamma + (1-e^2)\cos^2\gamma)$$

$$= 4\sin^2\alpha(\sin^2\gamma + \cos^2\gamma - e^2\cos^2\gamma)$$

$$= 4\sin^2\alpha(1 - \underbrace{e^2\cos^2\frac{u_1+u_0}{2}}_{\cos^2\beta})$$

$$|\vec{r}_1 - \vec{r}_0|^2 = 4\sin^2\alpha\sin^2\beta$$

$$\boxed{Q_0Q_1 = 2\sin\alpha\sin\beta}$$

14.2 $\ddot{\vec{r}} = -\frac{\mu \vec{r}}{r^3}$ (14.2)

$(r\dot{r})^2 + C^2 = 2(\mu r + hr^2)$ $\quad C=0 \quad \mu = G(m_1+m_2)$

$(r\dot{r})^2 = 2(\mu + hr) \qquad r \to 0 \quad r\dot{r}^2 \to 2\mu$
$\qquad t \to t_1$

Show: $|\dot{r}| r^{1/2} \to \sqrt{2\mu}$

$r|t-t_1| \to (9/2\,\mu)^{1/3}$ as $t \to t_1$

$$r\dot{r}^2 = 2\mu + \cancel{2hr}^{\,0}$$

$$\sqrt{r\dot{r}^2} \to \sqrt{2\mu}$$

$$|\dot{r}| r^{1/2} \to \sqrt{2\mu} \quad \checkmark$$

$$\frac{dr}{dt} = -\sqrt{\frac{2\mu}{r}}$$

$$\int r^{1/2}\,dr = \int \sqrt{2\mu}\,dt$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu}\, t \Big|_{t}^{t_1}$$

$$\frac{2}{3} r^{3/2} = \sqrt{2\mu}(t-t_1)$$

$$r^{3/2} = \frac{3}{2}\sqrt{2\mu}(t-t_1)$$

$$r = \left[\frac{3}{2}\sqrt{2\mu}(t-t_1)\right]^{2/3}$$

$$r = \left(\frac{9}{2}\mu\right)^{1/3} (t-t_1)^{2/3}$$

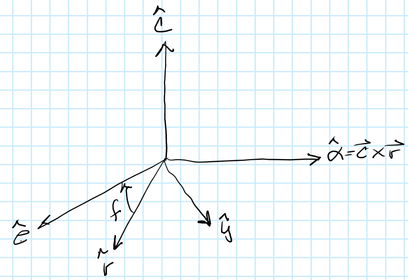
$$\frac{r}{(t-t_1)^{2/3}} = \left(\frac{9}{2}\mu\right)^{1/3}$$

$$\boxed{r(t-t_1)^{-2/3} \to \left(\frac{9}{2}\mu\right)^{1/3} \quad t \to t_1}$$

16.1 $\vec{F} = \frac{F_c}{c}\vec{c} + \frac{F_r}{r}\vec{r} + \frac{F_\alpha}{\alpha}\vec{\alpha}$

16.2 $\dot{\vec{c}} = r \times \left(-\frac{\mu \vec{r}}{r^3} + \vec{F}\right) = \vec{r} \times \vec{F}$

16.7 $\mu \dot{\vec{e}} = \vec{F} \times \vec{c} - Ar^2\vec{F} + \left[(A(\vec{F}\cdot\vec{r}) + \frac{\vec{F}\cdot\vec{\alpha}}{r^2}\right]\vec{r}$



$$\dot{\vec{c}} = r\frac{F_\alpha \vec{c}}{c} - \frac{F_c \vec{\alpha}}{c}$$

$$A = \frac{\mu e}{rc}\sin f$$

$\vec{r} \times \vec{r} = 0 \qquad \vec{c} \times \vec{c} = 0$

$\vec{r} \times \vec{c} = -\alpha \qquad \vec{\alpha} \times \vec{c} = \frac{c\vec{r}}{r}$

$\vec{r} \times \vec{\alpha} = \frac{r\vec{c}}{c}$

$\alpha = rc$

$$\dot{\vec{c}} = \vec{r} \times \left(\frac{F_c \vec{c}}{c} + \frac{F_r \vec{r}}{r} + \frac{F_\alpha \vec{\alpha}}{\alpha}\right)$$

$$= -\frac{F_c \vec{\alpha}}{c} + \frac{F_\alpha \vec{c}}{c} r$$

$$\boxed{\dot{\vec{c}} = -\frac{F_c \vec{\alpha}}{c} + r\frac{F_\alpha \vec{c}}{c}}$$

$$\mu\dot{\vec{e}} = \left(\frac{F_c\vec{c}}{c} + \frac{F_r\vec{r}}{r} + \frac{F_\alpha\vec{\alpha}}{\alpha}\right) \times \vec{c} - Ar^2\left(\frac{F_c\vec{c}}{c} + \frac{F_r\vec{r}}{r} + \frac{F_\alpha\vec{\alpha}}{\alpha}\right) + \left(AF_r r + \frac{F_\alpha}{r^2}\right)\cdot\vec{r}$$

$$= \underbrace{-\frac{F_r\vec{\alpha}}{r}}\ \underbrace{+\frac{cF_\alpha\vec{r}}{r}}\ \underbrace{-Ar^2\frac{F_c\vec{c}}{c}}\ \underbrace{-Ar^2\frac{F_r\vec{r}}{r}}\ \underbrace{-Ar^2\frac{F_\alpha\vec{\alpha}}{\alpha}} + \left(ArF_r + \frac{\alpha F_\alpha}{r^2}\right)\vec{r}$$

$$= -Ar^2\frac{F_c}{c}\vec{c} - \left(Ar^2\frac{F_\alpha}{rc} + \frac{F_r}{r}\right)\vec{\alpha} + \left(\frac{cF_\alpha}{r} - \cancel{AF_r} + \cancel{AF_r} + \frac{rcF_\alpha}{r^2}\right)\vec{r}$$

$$\boxed{\mu\dot{\vec{e}} = -Ar^2\frac{F_c}{c}\vec{c} + 2\frac{cF_\alpha}{r}\vec{r} - \left(Ar\frac{F_\alpha}{c} + \frac{F_r}{r}\right)\vec{\alpha}}$$

Hwk 1 Prob 2

Friday, January 23, 2026 10:20 PM

2) prove identities

Linear momentum:

$$F_{ij} = \frac{Gm_im_j}{r_{ij}^3}\vec{r}_{ji} = -\frac{Gm_im_j}{r_{ji}^3}\vec{r}_{ij}$$

$$\sum_{i=1}^{N} F_i = \sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \frac{Gm_im_j}{\|r_{ij}\|^3}\vec{r}_{ji}$$

$$= \sum\left(\frac{m_im_j}{r_{ij}^3}\vec{r}_{ji} + \frac{m_im_j}{r_{ji}^3}\vec{r}_{ij}\right)$$

$$= \sum \frac{m_im_j}{r_{ij}^3}\left(\vec{r}_{ji} + \vec{r}_{ij}\right) = 0$$

so $\underbrace{\sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \frac{m_im_j}{r_{ij}^3}\vec{r}_{ji} = 0}$

angular momentum

$r_{ij} = r_{ji}$

$$X = \sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \frac{m_im_j}{r_{ij}^3}\vec{r}_i \times \vec{r}_{ji} = \sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \frac{m_im_j}{r_{ij}^3}\vec{r}_j \times \vec{r}_{ij}$$

$$2X = \sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N}\left(\frac{m_im_j}{r_{ij}^3}\vec{r}_i \times \vec{r}_{ji} + \frac{m_im_j}{r_{ij}^3}\vec{r}_j \times \vec{r}_{ij}\right)$$

$$\sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \frac{m_im_j}{r_{ij}^3}\left(\vec{r}_i \times \vec{r}_{ji} + \vec{r}_j \times \vec{r}_{ij}\right)$$

$$\sum_{i=1}^{N}\sum_{j=1}^{N} \frac{m_im_j}{r_{ij}^3}\left(\vec{r}_i \times (\vec{r}_i - \vec{r}_j) + \vec{r}_j \times (\vec{r}_j - \vec{r}_i)\right)$$

$$(-r_{ij} - r_{ji}) = 0$$

$2X = 0$

$X = 0$ so $\underbrace{\sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \frac{m_im_j}{r_{ij}^3}\vec{r}_i \times \vec{r}_{ji} = 0}$

energy $\quad r_{ij} = r_{ji}$

$$\sum_{i=1}^{N} \sum_{\substack{j=1 \\ i \neq j}}^{N} \frac{m_i m_j}{r_{ij}} = 2 \sum_{i<j} \frac{m_i m_j}{r_{ij}}$$

$$U = -\frac{G}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ i \neq j}}^{N} \frac{m_i m_j}{r_{ij}} = -\frac{G}{2} \left[2 \sum_{1 \le i < j \le N} \frac{m_i m_j}{r_{ij}} \right]$$

$$\boxed{U = -G \sum_{1 \le i < j \le N} \frac{m_i m_j}{r_{ij}}}$$

total energy

$$T = \frac{1}{2} \sum_{i=1}^{N} m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$$

$$\frac{dT}{dt} = \frac{1}{2} \sum_{i=1}^{N} m_i \left(\ddot{\vec{r}}_i \cdot \dot{\vec{r}}_i + \dot{\vec{r}}_i \cdot \ddot{\vec{r}}_i \right)$$

$$\frac{dT}{dt} = \sum_{i=1}^{N} m_i \dot{\vec{r}}_i \ddot{\vec{r}}_i$$

$$\vec{F} = m_i \ddot{\vec{r}}_i = \sum_{j=1, j \neq i}^{N} \frac{G m_i m_j}{r_{ij}^3} \vec{r}_{ji}$$

$$\frac{dT}{dt} = \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji}$$

$$\frac{dU}{dt} = -\frac{G}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \frac{d}{dt} \left(\frac{m_i m_j}{\|\vec{r}_{ij}\|} \right)$$

$$= -\frac{G}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} m_i m_j \left(\frac{-1}{\|\vec{r}_{ij}\|^2} \right) \frac{d\|\vec{r}_{ij}\|}{dt} \rightarrow \sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}}$$

$$= +\frac{G}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} m_i m_j \frac{1}{\|\vec{r}_{ij}\|^2} \left(\frac{1}{2\sqrt{\vec{r}_{ij} \cdot \vec{r}_{ij}}} \right) \frac{d}{dt} (\vec{r}_{ij} \cdot \vec{r}_{ij})$$

$$= \frac{G}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ i \neq j}}^{N} \frac{m_i m_j}{r_{ij}^3} (\dot{\vec{r}}_i - \dot{\vec{r}}_j) \cdot \vec{r}_{ij}$$

$$\frac{dU}{dt} = G \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \frac{m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ij}$$

$$\frac{d(T+U)}{dt} = \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji}$$

$$\sum^{N} \sum^{N} G m_i m_j \ldots$$

$$\frac{\quad}{dt} \quad \sum_{i=1} \sum_{\substack{j=1 \\ j \neq i}} \frac{\quad}{r_{ij}^3} \quad \cdots$$

$$- \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j \neq i}}^{N} \frac{G m_i m_j}{r_{ij}^3} \dot{\vec{r}}_i \cdot \vec{r}_{ji} = 0$$

$$\boxed{\frac{d(T+U)}{dt} = 0}$$

Hwk 1 Prob3

Friday, January 23, 2026 10:21 PM

3) $U = -G\sum_{1\le i<j\le N} \frac{m_i m_j}{\|\vec{r}_{ij}\|}$

$U = -G\sum_{1\le i<j\le N} \frac{m_i m_j}{\|\alpha \vec{r}_{ij}\|} = -\frac{G}{\alpha}\sum \frac{m_i m_j}{\|\vec{r}_{ij}\|} = \boxed{-G\alpha^{-1}\sum_{1\le i<j\le N} \frac{m_i m_j}{\|\vec{r}_{ij}\|} \Rightarrow n=-1}$

$\frac{\partial U}{\partial \vec{r}_i} = -G\sum_{1\le i<j\le N} \frac{m_i m_j}{r_{ij}^3} \vec{r}_i \cdot \vec{r}_{ij}$

$\vec{r}_i \frac{\partial U}{\partial \vec{r}_i} = G\sum_{1\le i<j\le N} \frac{m_i m_j}{r_{ij}^{\cancel{3}}} \underbrace{\vec{r}_i \cdot \vec{r}_i}_{\cancel{r^2}} \cdot \vec{r}_{ij}$

$\boxed{\vec{r}_i \frac{\partial U}{\partial \vec{r}_i} = G\sum_{1\le i<j\le N} \frac{m_i m_j}{r_{ij}} \vec{r}_{ij} = -U}$

$T = \frac{1}{2}\sum_{i=1}^{N} m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i$

$T = \frac{1}{2}\sum_{i=1}^{N} m_i \alpha\dot{\vec{r}}_i \cdot \alpha\dot{\vec{r}}_i = \boxed{\alpha^2\left(\frac{1}{2}\sum_{i=1}^{N} m_i \dot{\vec{r}}_i \cdot \dot{\vec{r}}_i\right) \Rightarrow n=2}$

$\frac{\partial T}{\partial \dot{\vec{r}}_i} = \sum_{i=1}^{N} m_i \dot{\vec{r}}$

$\boxed{\dot{\vec{r}}_i \frac{\partial T}{\partial \dot{\vec{r}}_i} = \sum_{i=1}^{N} m_i \dot{\vec{r}} \cdot \dot{\vec{r}} = 2T}$

$I_p = \sum_{i=1}^{N} m_i \vec{r}_i \cdot \vec{r}_i$

$I_p = \sum_{i=1}^{N} m_i \alpha\vec{r}_i \cdot \alpha\vec{r}_i = \boxed{\alpha^2 r_i \sum_{i=1}^{N} m_i \vec{r}_i \cdot \vec{r}_i \Rightarrow N=2}$

$\frac{\partial I_p}{\partial \vec{r}_i} = 2\sum_{i=1}^{N} m_i \vec{r}_i$

$\boxed{\vec{r} \cdot \frac{\partial I_p}{\partial \vec{r}_i} = 2\sum_{i=1}^{N} m_i \vec{r}_i \cdot \vec{r}_i = 2I_p}$

$I_p U^2 = \sum_{i=1}^{N} m_i \vec{r}_i \cdot \vec{r}_i \cdot \left[G\sum_{1\le i<j\le N} \frac{m_i m_j}{r_{ij}}\right]^2$

$\boxed{= \sum_{i=1}^{N} m_i \cancel{\alpha}\vec{r}_i \cdot \cancel{\alpha}\vec{r}_i \left(G\sum_{1\le i<j\le N} \frac{m_i m_j}{\cancel{\alpha} r_{ij}}\right)^2 \Rightarrow n=0}$

for homogeneous functions order product of functions equals sum of order $(2 + 2(-1) = 0)$ so order $= 0$

Hwk 1 Prob 4

Friday, January 23, 2026 10:22 PM

4) $\vec{H} = \sum_{i=1}^{N} m_i \vec{r}_i \times \dot{\vec{r}}_i$ $\quad H^2 \leq 2T I_p$ $\quad a = \vec{r} \quad b = \dot{\vec{r}}$

$\vec{H} = \sum m_i \, a \times b$

$\vec{H}^2 = \sum m_i \, a \times b \sum m_i \, a \times b$ $\quad T = \frac{1}{2} \sum m_i a^2$ $\quad A = 2T$

$= \sum m_i^2 (a \times b)(a \times b)$ $\quad I_p = \sum m_i b^2$ $\quad B = I_p$

$H^2 = \sum m_i^2 \left[a^2 b^2 - (ab)(ba) \right]$

$\boxed{H^2 \leq 2T I_p}$

Hwk 1 Prob 5

Friday, January 23, 2026 10:22 PM

5) Lagrange Identity

$$\left(\sum_{k=1}^{N} a_k^2\right)\left(\sum_{k=1}^{N} b_k^2\right) - \left(\sum_{k=1}^{N} a_k b_k\right)^2 = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}(a_i b_j - a_j b_i)^2$$

$$I_p = \sum_{i=1}^{N} m_i r_i^2 = \sum_{i=1}^{N} m_i \vec{r}_i \cdot \vec{r}_i)$$

$$r_{ij} = r_i - r_j$$

$$\frac{1}{2M} \sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j (\vec{r}_i - \vec{r}_j)^2$$

$$\frac{1}{2M} \sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j (r_i^2 - 2\vec{r}_i \vec{r}_j + r_j^2)$$

$$\frac{1}{2M}\left(\sum_{i=1}^{N} m_i r_i^2 \underbrace{\sum_{j=1}^{N} m_j}_{M} - 2\sum_{i=1}^{N}\sum_{j=1}^{N} \vec{r}_i \vec{r}_j + \sum_{i=1}^{N} m_i \sum_{j=1}^{N} m_j r_j^2\right)$$

$$\frac{1}{2}\underbrace{\sum_{i=1}^{N} m_i r_i^2}_{I_p} - \frac{1}{M}\underbrace{\sum_{i=1}^{N} m_i \vec{r}_i}_{MR_c} \cdot \underbrace{\sum_{j=1}^{N} m_j \vec{r}_j}_{MR_c} + \frac{1}{2M}\underbrace{\sum_{i=1}^{N} m_i}_{M} \underbrace{\sum_{j=1}^{N} m_j r_j^2}_{I_p}$$

$$\frac{1}{2} I_p + \frac{1}{2} I_p - M \vec{R}_c \cdot \vec{R}_c$$

$$\frac{1}{2M}\sum_{i}^{N}\sum_{j}^{N} m_i m_j \vec{r}_{ij} \cdot \vec{r}_{ij} = I_p - M\vec{R}_c \cdot \vec{R}_c$$

$$\boxed{I_p = \frac{1}{2M}\sum_{i}^{N}\sum_{j}^{N} m_i m_j \vec{r}_{ij} \cdot \vec{r}_{ij} + M\vec{R}_c \cdot \vec{R}_c}$$

$$T = \frac{1}{2}\sum_{i=1}^{N} m_i v_i^2$$

$$\frac{1}{2M} \sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j v_{ij}^2 = \frac{1}{2M}\sum_{i}^{N}\sum_{j=1}^{N} m_i m_j (v_i - v_j)^2$$

$$= \frac{1}{2M}\sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j (v_i^2 - 2\vec{v}_i \vec{v}_j + v_j^2)$$

$$= \frac{1}{2M}\left(\sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j v_i^2 - \sum\sum 2 m_i m_j \vec{v}_i \vec{v}_j + \sum\sum m_i m_j v_j^2\right)$$

$$= \frac{1}{2M}\left(\sum_{i=1}^{N} m_i v_i^2 \sum m_j \; 2\sum_{i=1}^{N} m_i \vec{v}_i \sum_{j=1}^{N} m_j \vec{v}_j + \sum_{i=1}^{N} m_i \sum_{j=1}^{N} m_j v_j^2\right)$$

$$= \frac{1}{2M}\Big(\underbrace{\sum_{i=1}^{N} m_i v_i^2}_{T} \underbrace{\sum m_j}_{M} \; 2\underbrace{\sum_{i=1}^{N} m_i \vec{v}_i}_{v_c} \underbrace{\sum_{j=1}^{N} m_j \vec{v}_j}_{v_c} + \underbrace{\sum_{i=1}^{N} m_i}_{M} \underbrace{\sum_{j=1}^{N} m_j v_j^2}_{T}\Big)$$

$$= \frac{1}{2}\left(2T - M\,\vec{v}_c \cdot \vec{v}_c\right)$$

$$\boxed{T = \frac{1}{2M}\sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j v_{ij}^2 + \frac{1}{2}M\,\vec{v}_c \cdot \vec{v}_c}$$

Hwk 1 Prob 6.

Friday, January 23, 2026 10:23 PM

6) show: $\displaystyle\sum_{i=1}^{N-1}\sum_{\substack{j=1\\ j\neq i}}^{N} f_{ij} = \frac{1}{2}\sum_{1\le i<j\le N} f_{ij}\ ;\ f_{ii}=0$

$$\sum_{i=1}^{N} m_i\tilde{r}_i\cdot\vec{v}_i = \frac{1}{M}\sum_{i=1}^{N}\sum_{j=i+1}^{N} m_i m_j \tilde{r}_{ij}\cdot\vec{v}_{ij}$$

$\tilde{r}_i\cdot\vec{v}_i$ @ $i=j$ $\tilde{r}_{ij}\cdot\vec{v}_{ij}=0$

$$\frac{1}{M}\Big[\frac{1}{2}\sum_{1\le i<j\le N} m_i m_j(\tilde{r}_j-\tilde{r}_i)\cdot(\vec{v}_j-\vec{v}_i)\Big]$$

$$\frac{1}{2M}\sum_{1\le i<j\le N} m_i m_j\big[\tilde{r}_j\cdot\vec{v}_j-\tilde{r}_j\cdot\vec{v}_i-\tilde{r}_i\cdot\vec{v}_j+\tilde{r}_i\vec{v}_i\big]$$

$$=\frac{1}{2M}\Big[\Big(\sum_{j=1}^{N} m_j\tilde{r}_j\cdot\vec{v}_j\Big)\Big(\sum_i m_i\Big)-\underbrace{\Big(\sum_{j=1}^{N} m_j\tilde{r}_j\Big)}_{M\tilde{R}_c}\cdot\underbrace{\Big(\sum_{i=1}^{N} m_i\vec{v}_i\Big)}_{M\vec{V}}-\underbrace{\Big(\sum_i^{N} m_i\tilde{r}_i\Big)}_{M\tilde{R}_c}\cdot\underbrace{\Big(\sum_{j=1}^{N} m_j\vec{v}_j\Big)}_{M\vec{V}}$$
$$+\Big(\sum_{i=1}^{N} m_i\tilde{r}_i\cdot\vec{v}_i\Big)\Big(\sum_{j=1}^{N} m_j\Big)\Big]$$

$$\sum_{j=1}^{N} m_j\tilde{r}_j\cdot\vec{v}_j=\sum m_i\tilde{r}_i\cdot\vec{v}_i$$

$$=\frac{1}{2M}\Big(2M\sum_{i=1}^{N} m_i\tilde{r}_i\cdot\vec{v}_i-2M\tilde{R}_c\cdot M\vec{V}\Big)$$

$$=\sum_{i=1}^{N} m_i\tilde{r}_i\cdot\vec{v}_i-\cancel{M\tilde{R}_c}^{\,0}\cdot\vec{V}\ \checkmark \qquad \text{@ COM } \tilde{R}_c=0$$

$$\boxed{\sum_{i=1}^{N} m_i\tilde{r}_i\cdot\vec{v}_i=\frac{1}{M}\sum_{i=1}^{N}\sum_{j=i+1}^{N} m_i m_j\tilde{r}_{ij}\cdot\vec{v}_{ij}}$$

show:

$$\sum_{i=1}^{N} m_i\big[r_i^2\,\overline{U}-\vec{r}_i\vec{r}_i\big]=\frac{1}{M}\sum_{i=1}^{N}\sum_{j=i+1}^{N} m_i m_j\big[r_{ij}^2\,\overline{U}-\vec{r}_{ij}\vec{r}_{ij}\big]$$

$$=\frac{1}{2M}\sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j\big[r_{ij}^2\,\overline{U}-\vec{r}_{ij}\vec{r}_{ij}\big]$$

$$=\frac{1}{2M}\sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j\big[(r_j^2-2\vec{r}_i\vec{r}_j+r_i^2)\overline{U}-(\vec{r}_j\vec{r}_j-\vec{r}_j\vec{r}_i-\vec{r}_i\vec{r}_j+\vec{r}_i\vec{r}_i)\big]$$

$$=\frac{1}{2M}\sum_{i=1}^{N}\sum_{j=1}^{N}\underbrace{m_i m_j}_{M}\Big\{\underbrace{\big[r_i^2\,\overline{U}-\vec{r}_i\vec{r}_i\big]+\big[r_j^2\,\overline{U}-\vec{r}_j\vec{r}_j\big]}_{2r_i^2\overline{U}-\vec{r}_i\vec{r}_i}-2(\vec{r}_i\cdot\vec{r}_j)\overline{U}+\vec{r}_i\vec{r}_j+\vec{r}_j\vec{r}_i\Big\}$$

$$=\sum_{i=1}^{N} r_i^2\,\overline{U}-\vec{r}_i\vec{r}_i+\frac{1}{M}\sum_{i=1}^{N}\sum_{j=1}^{N} m_i m_j\,{-2}(\vec{r}_i\cdot\vec{r}_j)\overline{U}+\vec{r}_i\vec{r}_j+\vec{r}_j\vec{r}_i\Big]$$

$$=\sum_{i=1}^{N} r_i^2\,\overline{U}-\vec{r}_i\vec{r}_i-\frac{1}{M}\Big[\Big(\sum_{i=1}^{N} m_i\vec{r}_i\sum_{j=1}^{N} m_j\vec{r}_j\Big)\overline{U}+\sum_{i=1}^{N} m_i\vec{r}_i\sum_{j=1}^{N} m_2\vec{r}_j+\sum_{j=1}^{N} m_j r_j\sum_{i=1}^{N} m_i r_i$$

$$=\sum_{i=1}^{N} r_i^2\,\overline{U}-\vec{r}_i\vec{r}_i-\frac{1}{M}\cancel{M\overline{R}_c}^{\,0}\,M\overline{R}_c\,\overline{U}\quad+2\cancel{M}\vec{R}_c^{\,0}\vec{R}_c$$

@ COM $R_c=0$

$$\boxed{\sum_{i=1}^{N} r_i^2\,\overline{U}-\vec{r}_i\vec{r}_i=\frac{1}{M}\sum_{i=1}^{N}\sum_{j=i+1}^{N} m_i m_j\big[r_{ij}^2\,\overline{U}-\vec{r}_{ij}\cdot\vec{r}_{ij}\big]}$$

Hwk 1 Prob 7

Friday, January 23, 2026 10:23 PM

7) $M_1 = m_1 \quad M_2 = m_1 + m_2 \quad M_3 = m_1 + m_2 + m_3$

$\vec{R}_1 = \vec{r}_2 - \vec{r}_1 \quad \vec{\rho} = \vec{R}_2 = \vec{r}_3 - \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2} \qquad R_{cm} = \frac{m_1 r_1 + m_2 r_2 + m_3 r_3}{M}$

$T = \frac{1}{2}\sum_{i=1}^{3} m_i \dot{r}_i^2$ define $M_i = \sum_{j=1}^{N} m_j$; $\vec{R}_N \equiv$ system center of mass

$T = \frac{1}{2}\sum_{i=1}^{N-1} M_{i-1}\frac{M_{i-1} m_i}{M_i}\dot{\vec{R}}_i \cdot \dot{\vec{R}}_i + \underbrace{\frac{1}{2} M_N \dot{\vec{R}}_N \cdot \dot{\vec{R}}_N}_{\text{KE of CoM}}$

$T = \frac{1}{2}\frac{m_1 m_2}{m_1+m_2}\dot{\vec{R}}\cdot\dot{\vec{R}} + \frac{1}{2}\frac{m_3(m_1+m_2)}{m_1+m_2+m_3}\dot{\vec{\rho}}\cdot\dot{\vec{\rho}}$

$\mu_1 = \frac{m_1 m_2}{M_2} \qquad \mu_2 = \frac{M_2 m_3}{M_3}$

COM: $\vec{R}_1 = \vec{r}_2 - \vec{R}_1^c = \vec{r}_2 - \vec{r}_1$

$\vec{R}_2 = \vec{r}_3 - R_2^c = r_3 - \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2}$

$\vec{r}_3 = \vec{R}_2^c + \vec{\rho}$

$\vec{r}_2 = \vec{R}_2^c + \frac{m_1}{M_2}\vec{R}_1$

$\vec{r}_1 = \vec{R}_2^c - \frac{m_2}{M_2}\vec{R}_1$

$U = -\frac{Gm_1m_2}{|\vec{R}|} - \frac{Gm_1m_3}{|\vec{\rho} + \frac{m_2}{M_2}\vec{R}|} - \frac{Gm_2m_3}{|\vec{\rho} - \frac{m_1}{M_2}\vec{R}|}$

$\mathcal{L} = T - U$

$\frac{d}{dt}\frac{\partial T}{\partial \dot{\vec{R}}_j} = -\frac{\partial U}{\partial \vec{R}_j}$

$\frac{d}{dt}\frac{\partial T}{\partial \dot{\vec{R}}} = \frac{m_1 m_2}{M_2}\ddot{\vec{R}}$

$\frac{d}{dt}\frac{\partial T}{\partial \dot{\vec{\rho}}} = \frac{m_3 M_2}{M_3}\ddot{\vec{\rho}}$

$-\frac{\partial}{\partial \vec{R}}\left(-\frac{Gm_1m_2}{R}\right) = -Gm_1m_2\frac{\vec{R}}{R^3}$

$-\frac{\partial}{\partial R}\left(\frac{-Gm_1m_3}{\vec{\rho} + \frac{m_2}{M_2}\vec{R}}\right) = \frac{-Gm_1m_3}{|\vec{\rho} + \frac{m_2}{M_2}\vec{R}|^3}\frac{m_2}{M_2}\left(\vec{\rho} + \frac{m_2}{M_2}\vec{R}\right)$

$-\frac{\partial}{\partial R}\left(\frac{-Gm_2m_3}{|\vec{\rho} - \frac{m_1}{M_2}\vec{R}|}\right) = \frac{-Gm_2m_3}{|\vec{\rho} - \frac{m_1}{M_2}\vec{R}|^3}\left(-\frac{m_1}{M_2}\right) = \frac{Gm_1m_2m_3\left(\vec{\rho} - \frac{m_1}{M_2}\vec{R}\right)}{M_2|\vec{\rho} - \frac{m_1}{M_2}\vec{R}|^3}$

$-\frac{\partial U}{\partial R} = \boxed{-\frac{Gm_1m_2}{R^3}\vec{R} - \frac{Gm_1m_2m_3}{M_2}\left[\frac{\vec{\rho} + \frac{m_2}{M_2}\vec{R}}{|\vec{\rho} + \frac{m_2}{M_2}\vec{R}|^3} - \frac{\vec{\rho} - \frac{m_1}{M_2}\vec{R}}{|\vec{\rho} - \frac{m_1}{M_2}\vec{R}|^3}\right] = \frac{m_1m_2}{M_2}\ddot{\vec{R}}}$

$$-\frac{\partial U}{\partial R} = \boxed{-\frac{Gm_1m_2}{R^3}\vec{R} - \frac{Gm_1m_2m_3}{M_2}\left[\frac{\bar{\rho} + \frac{m_2}{M_2}\vec{R}}{\left|\bar{\rho} + \frac{m_2}{M_2}\vec{R}\right|^3} - \frac{\bar{\rho} - \frac{m_1}{M_2}\vec{R}}{\left|\bar{\rho} - \frac{m_1}{M_2}\vec{R}\right|^3}\right] = \frac{m_1 m_2 \overline{R}}{M_2}}$$

$$-\frac{\partial U}{\partial \rho}\left(\frac{-Gm_1m_3}{\bar{\rho} + \frac{m_2}{M_2}\vec{R}}\right) = -Gm_1m_3\,\frac{\bar{\rho} + \frac{m_2}{M_2}\vec{R}}{\left|\bar{\rho} + \frac{m_2}{M_2}\vec{R}\right|^3}$$

$$-\frac{\partial U}{\partial \rho}\left(\frac{-Gm_2m_3}{\bar{\rho} - \frac{m_1}{M_2}\vec{R}}\right) = -Gm_2m_3\,\frac{\bar{\rho} - \frac{m_1}{M_2}\vec{R}}{\left|\bar{\rho} - \frac{m_1}{M_2}\vec{R}\right|^3}$$

$$\frac{\partial U}{\partial \rho} = \boxed{\frac{m_3M_2}{M_3}\,\ddot{\bar{\rho}} = -Gm_1m_3\,\frac{\bar{\rho} + \frac{m_2}{M_2}\vec{R}}{\left|\bar{\rho} + \frac{m_2}{M_2}\vec{R}\right|^3} - Gm_2m_3\,\frac{\bar{\rho} - \frac{m_1}{M_2}\vec{R}}{\left|\bar{\rho} - \frac{m_1}{M_2}\vec{R}\right|^3}}$$

$N=4$ Jacobi

$M_4 = m_1 + m_2 + m_3 + m_4$

$$\boxed{\vec{R}_3 = \vec{r}_4 + \vec{R}_3^{c} = \vec{r}_4 - \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{M_3}}$$

$N=5$ Jacobi

$M_5 = M_4 + m_5$

$$\boxed{\vec{R}_4 = \vec{r}_5 + R_4^{c} = \vec{r}_5 - \frac{m_1r_1 + m_2r_2 + m_3r_3 + m_4r_4}{M_4}}$$